STRAVA FITNESS DATA ANALYSIS

Complete Data Analytics Project

Project Completed As Part of

Data Analyst Internship

Prepared By

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Tools Used:

Python, PostgreSQL, Power BI

Smart Fitness Tracker Data Analysis

Understanding User Behaviour Patterns Using Python, SQL and Power BI

1. Project Summary

This project analyzes smart fitness tracker data to understand user behavior patterns related to physical activity, sleep habits, calorie expenditure, and BMI. The workflow involved Python for data preprocessing and exploratory data analysis (EDA), PostgreSQL for structured querying, and Power BI for interactive dashboards. The insights help understand how users interact with fitness trackers and how such data can support improved marketing strategies and personalized health recommendations.

2. Tools and Technologies Used

Python (Pandas, Matplotlib, Seaborn) – Data preprocessing and exploratory data analysis

PostgreSQL – Structured querying and relational data analysis

Power BI – Interactive dashboard visualization

Jupyter Notebook – Data analysis environment

2. Business Objective

The objective of this project is to analyze smart fitness tracker data to understand user behaviour patterns including activity levels, sleep habits, calorie expenditure and BMI. These insights help organizations design better health features, improve user engagement and build targeted marketing strategies.

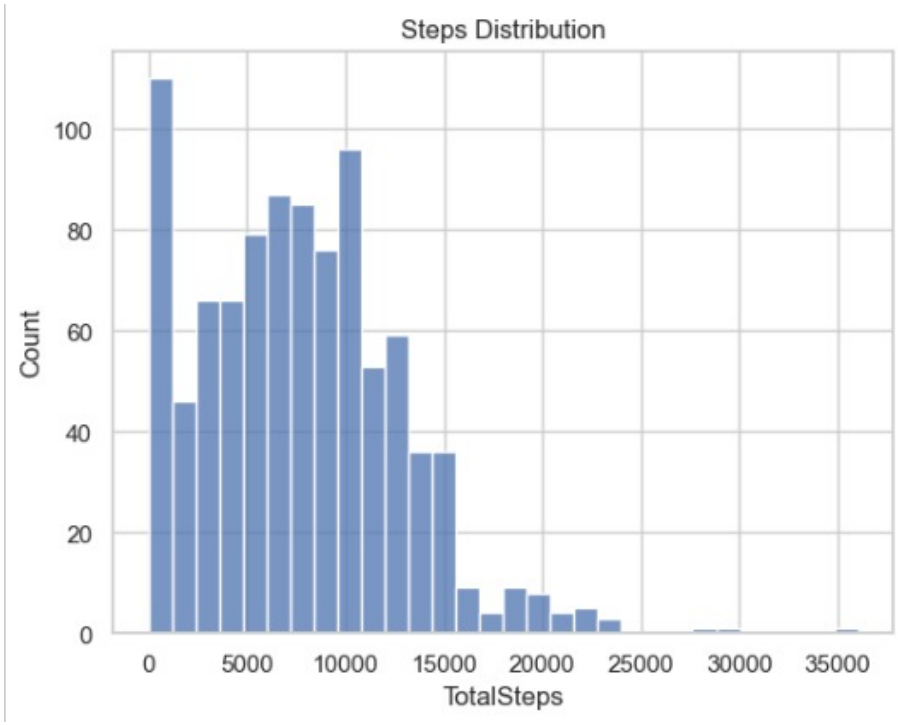
3. Dataset Overview

The dataset contains information collected from wearable fitness trackers. It includes multiple variables such as daily steps, hourly activity, calories burned, sleep duration, heart rate, weight and BMI. The dataset is divided into multiple tables including daily_activity,

hourly_steps, hourly_calories, sleep_day, minute_sleep, heartrate, weight_bmi and merged_fitness.

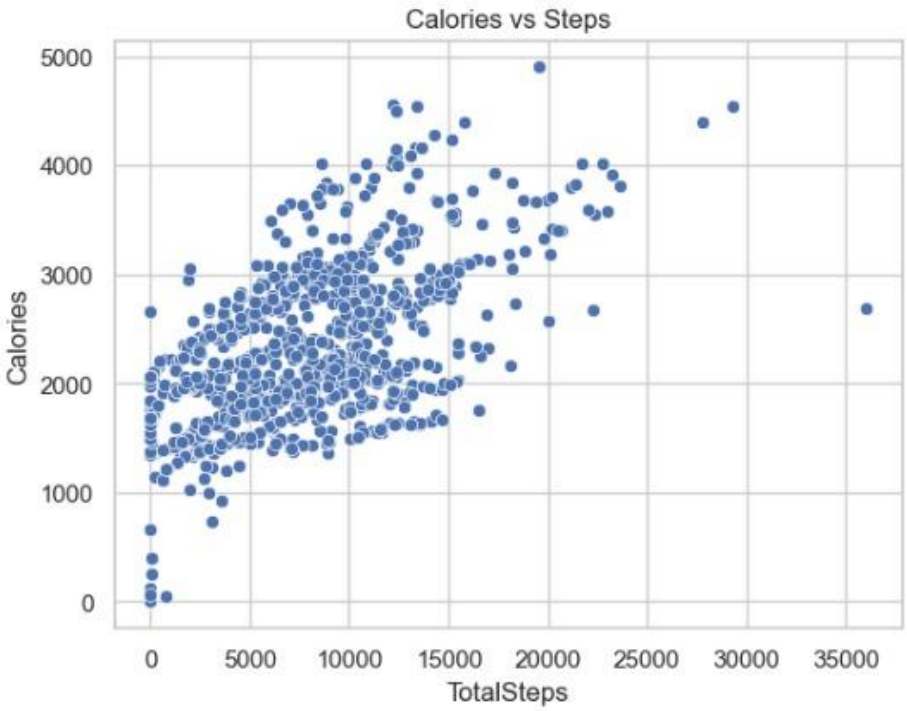
5. Python Exploratory Data Analysis (EDA)

5.1 Steps Distribution



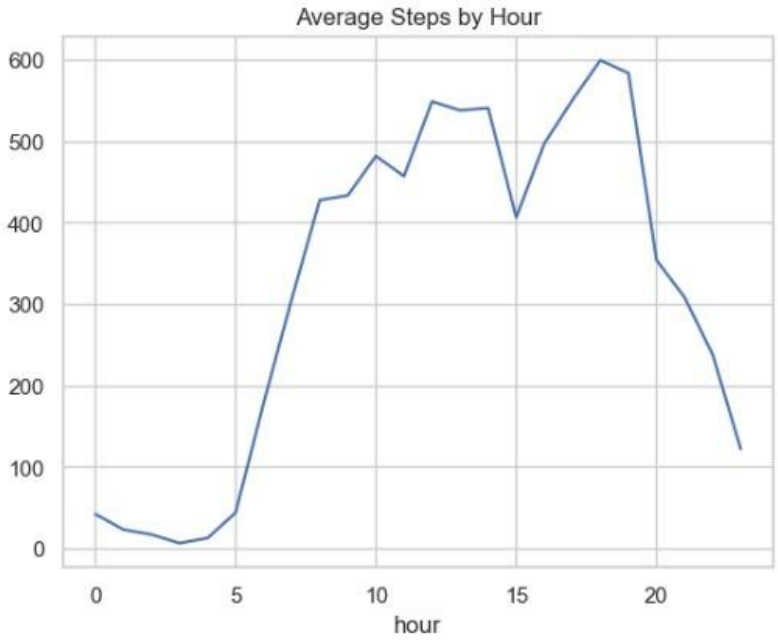
Insights: The analysis shows that most users record between 5,000 and 10,000 steps per day, indicating moderate activity levels.

5.2 Calories vs Steps



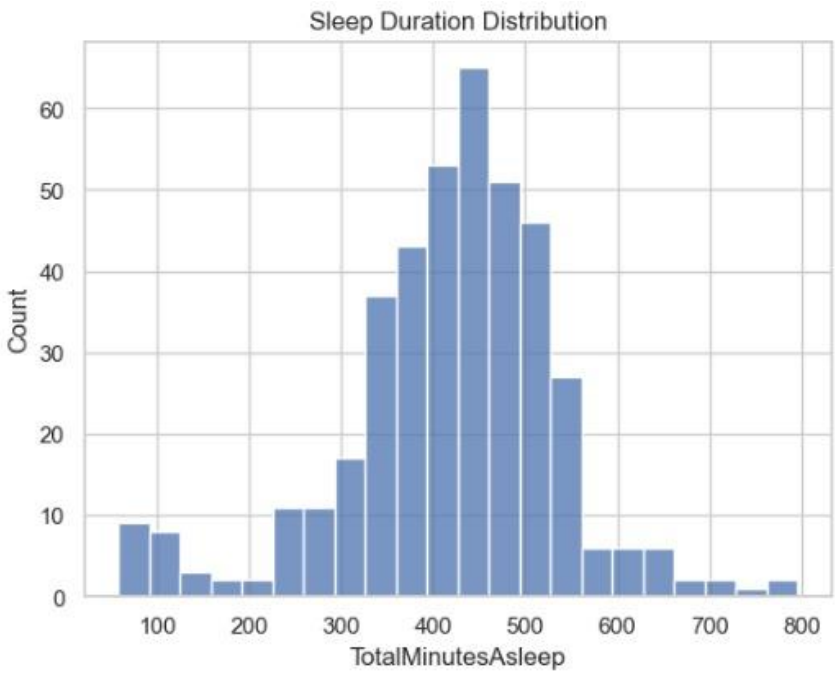
Insights: It shows a positive correlation between the two variables, indicating that higher physical activity results in higher calorie expenditure.

5.3 Average Steps by Hour



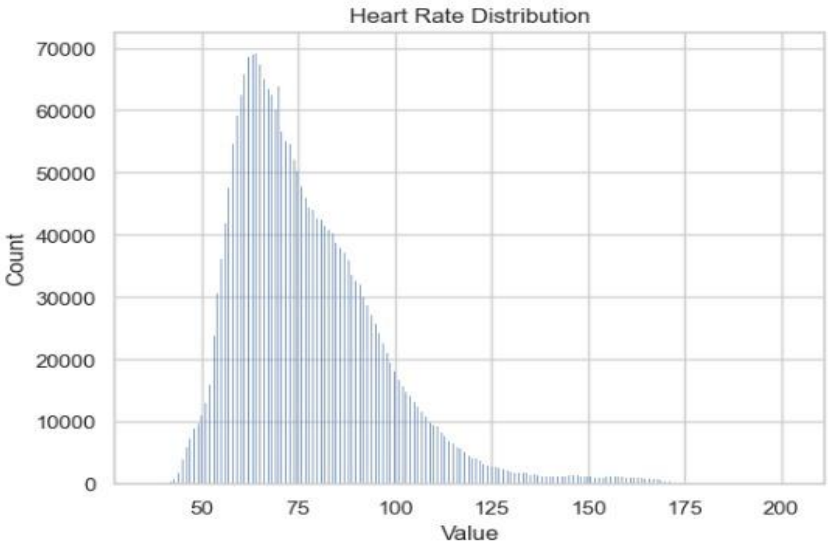
Insights: The data shows that user activity increases throughout the morning and afternoon and peaks in the evening before declining at night.

5.4 Sleep Duration Distribution



Insights: It shows that most users sleep between 6 and 8 hours per night, which aligns with recommended healthy sleep durations.

5.5 Heart Rate Distribution



Insights: The histogram indicates that most readings fall between 60 and 90 beats per minute, which is considered a normal resting heart rate range.

6. SQL Data Analysis (PostgreSQL)

SQL Query 1: Steps Frequency

```
SELECT totalsteps, COUNT(*) AS frequency

FROM daily_activity

GROUP BY totalsteps

ORDER BY totalsteps;
```

Output:

	totalsteps integer	frequency bigint
1	0	77
2	4	1
3	8	1
4	9	1
5	16	1
6	17	1
7	29	1
8	31	1
9	42	1
10	44	1

This query calculates how frequently different step counts occur in the dataset, helping identify common daily activity levels.

SQL Query 2: Average Steps by Hour

```
SELECT hour, AVG(steptotal) AS avg_steps

FROM hourly_steps

GROUP BY hour

ORDER BY hour;
```

Output:

	hour integer	avg_steps numeric
1	0	42.1884368308351178
2	1	23.1028938906752412
3	2	17.1103965702036442
4	3	6.4265809217577706
5	4	12.6995708154506438
6	5	43.8690987124463519
7	6	178.5080558539205156
8	7	306.0494092373791622
9	8	427.5445757250268528
10	9	433.3018259935553169

This query analyzes average step counts for each hour of the day, showing peak activity periods.

SQL Query 3: Average Calories by Hour

```
SELECT hour, AVG(calories) AS avg_calories

FROM hourly_calories

GROUP BY hour

ORDER BY hour;
```

Output:

	hour integer	avg_calories numeric
1	0	71.8051391862955032
2	1	70.1650589496248660
3	2	69.1864951768488746
4	3	67.5380493033226152
5	4	68.2618025751072961
6	5	81.7081545064377682
7	6	86.9967776584317938
8	7	94.4779806659505908
9	8	103.3372717508055854
10	9	106.1428571428571429

This query evaluates how calorie burn varies throughout the day based on hourly activity.

SQL Query 4: BMI Category Distribution

```
SELECT bmi_category, COUNT(*) AS users

FROM weight_bmi

GROUP BY bmi_category;
```

Output:

	bmi_category text	users bigint
1	Overweight	32
2	Normal	34
3	Obese	1

This query counts the number of users in each BMI category, helping understand overall health distribution.

SQL Query 5: Steps vs Sleep

```
SELECT totalsteps, totalminutesasleep

FROM merged_fitness

WHERE totalminutesasleep IS NOT NULL;
```

Output:

	totalsteps integer	totalminutesasleep double precision
1	13162	327
2	10735	384
3	9762	412
4	12669	340
5	9705	700
6	15506	304
7	10544	360
8	9819	325
9	14371	361
10	10039	430

This query analyzes the relationship between daily activity levels and sleep duration.

7. Power BI Dashboard Analysis

Three dashboards were created in Power BI to visualize user engagement, sleep patterns and BMI insights. These dashboards present interactive visualizations that allow stakeholders to quickly understand trends and patterns.

Dashboard 1: User Engagement

Key Insights

- Average daily steps are approximately 7.6K
- Average calories burned are around 2.3K
- Most users maintain moderate physical activity levels

Visual Insights

- Steps Distribution shows most users fall between 5K–10K steps
- Steps vs Calories scatter plot confirms that higher activity leads to higher calorie burn
- Average Steps by Hour shows peak activity during afternoon and evening

Dashboard 2: Sleep Pattern Analysis

Key Insights

- Average sleep duration is approximately **7 hours**
- Majority of users maintain **good sleep efficiency**

Visual Insights

- Most users sleep between **6–8 hours**
- Sleep efficiency levels are generally **high**
- Physical activity appears to have **some relationship with sleep duration**

Dashboard 3: BMI and Weight Analysis

Key Insights

- Average BMI is approximately 25.19
- Most users fall within Normal or Overweight categories

Visual Insights

- BMI category distribution shows Normal weight as the largest group
- Weight and BMI show a positive correlation
- Most users fall in the 60–80 kg weight range

8. Conclusion

The project demonstrates how fitness tracker data can be used to understand user behaviour patterns related to activity, sleep and health indicators. Using Python, SQL and Power BI together allows efficient data analysis and interactive visualization.

9. Business Recommendations

Introduce step-based challenges to increase user engagement.

Provide personalized health recommendations using activity and BMI data.

Improve sleep tracking features to encourage better sleep habits.

Use targeted marketing campaigns for users with higher BMI.

Implement gamification strategies to motivate consistent physical activity.